```
import pandas as pd
import numpy as np
```

```
from google.colab import files

uploaded = files.upload()

filename = next(iter(uploaded))

df = pd.read_csv("Online_Retail.csv")

print("Dataset loaded successfully!")
```

Choose Files Online_Retail.csv

Online_Retail.csv(text/csv) - 47217062 bytes, last modified: 10/29/2026 - 100% done

```
Saving Online_Retail.csv to Online_Retail.csv
Dataset loaded successfully!
```

```
print("First 5 rows:")
display(df.head())

print("Dataset shape:")
print(df.shape)

print("Column names:")
print(df.columns.tolist())

print("Dataset information:")
df.info()
```

First 5 rows:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/01/2010 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/01/2010 08:26:00	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom

```
Dataset shape:
(541909, 8)
Column names:
['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate', 'UnitPrice', 'CustomerID', 'Country']
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    541909 non-null  object
 1   StockCode    541909 non-null  object
 2   Description  540455 non-null  object
 3   Quantity     541909 non-null  int64
 4   InvoiceDate  541909 non-null  object
```

```
print("Missing values in each column:")
print(df.isnull().sum())

print("\nMissing values percentage:")
print((df.isnull().sum() / len(df)) * 100)
```

```
Missing values in each column:
InvoiceNo           0
StockCode           0
Description      1454
Quantity            0
InvoiceDate         0
UnitPrice           0
CustomerID     135080
```

```
Country             0
dtype: int64

Missing values percentage:
InvoiceNo       0.000000
StockCode       0.000000
Description     0.268311
Quantity        0.000000
InvoiceDate     0.000000
UnitPrice       0.000000
CustomerID     24.926694
Country         0.000000
dtype: float64
```

```
df_clean = df.copy()

important_columns = ["InvoiceNo", "Quantity", "UnitPrice"]

df_clean = df_clean.dropna(subset=important_columns)

print("Shape before cleaning:", df.shape)
print("Shape after handling missing important values:", df_clean.shape)

print("\nRemaining missing values:")
print(df_clean.isnull().sum())
```

```
Shape before cleaning: (541909, 8)
Shape after handling missing important values: (541909, 8)

Remaining missing values:
InvoiceNo           0
StockCode           0
Description      1454
Quantity            0
InvoiceDate         0
UnitPrice           0
CustomerID     135080
Country             0
dtype: int64
```

```
print("Repeated InvoiceNo values:")
print(df_clean["InvoiceNo"].duplicated().sum())

print("\nExample of repeated invoices:")
display(df_clean[df_clean["InvoiceNo"].duplicated(keep=False)].head(10))
```

```
Repeated InvoiceNo values:
516009

Example of repeated invoices:
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/01/2010 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/01/2010 08:26:00	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom
5	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	12/01/2010 08:26:00	7.65	17850.0	United Kingdom
			GLASS STAR FROSTED T-LIGHT		12/01/2010			United

```
print("Exact duplicate rows before removal:")
print(df_clean.duplicated().sum())

df_clean = df_clean.drop_duplicates()

print("Exact duplicate rows after removal:")
print(df_clean.duplicated().sum())
```

```
Exact duplicate rows before removal:
5268
Exact duplicate rows after removal:
0
```

```
df_clean["UnitPrice"] = pd.to_numeric(
    df_clean["UnitPrice"], errors="coerce"
)

valid_prices = df_clean[df_clean["UnitPrice"] >= 0]

highest_price_product = valid_prices.loc[
    valid_prices["UnitPrice"].idxmax()
]

print("Highest-priced product:")
print("Product:", highest_price_product["Description"])
print("Unit Price:", highest_price_product["UnitPrice"])
```

```
Highest-priced product:
Product: Manual
Unit Price: 38970.0
```

```
df_clean["Quantity"] = pd.to_numeric(
    df_clean["Quantity"], errors="coerce"
)

df_clean["Revenue"] = (
    df_clean["Quantity"] * df_clean["UnitPrice"]
)

display(df_clean[["InvoiceNo", "Description", "Quantity", "UnitPrice", "Revenue"]].head())
```

	InvoiceNo	Description	Quantity	UnitPrice	Revenue
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	2.55	15.30
1	536365	WHITE METAL LANTERN	6	3.39	20.34
2	536365	CREAM CUPID HEARTS COAT HANGER	8	2.75	22.00
3	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	3.39	20.34
4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	3.39	20.34

```
country_revenue = df_clean.groupby("Country")["Revenue"].sum()

print("Revenue by country:")
print(country_revenue.sort_values(ascending=False))
```

```
Revenue by country:
Country
United Kingdom          8167128.184
Netherlands              284661.540
EIRE                     262993.380
Germany                  221509.470
France                   197317.110
Australia                137009.770
Switzerland               56363.050
Spain                     54756.030
Belgium                   40910.960
Sweden                    36585.410
Japan                     35340.620
Norway                    35163.460
Portugal                  29302.970
Finland                   22326.740
Channel Islands           20076.390
Denmark                   18768.140
Italy                     16890.510
Cyprus                    12858.760
Austria                   10154.320
Hong Kong                  9908.240
Singapore                  9120.390
Israel                     7901.970
Poland                     7213.140
Unspecified                4740.940
Greece                     4710.520
Iceland                    4310.000
Canada                     3666.380
```

```
Malta                     2505.470
United Arab Emirates      1902.280
USA                       1730.920
Lebanon                   1693.880
Lithuania                 1661.060
European Community        1291.750
Brazil                    1143.600
RSA                       1002.310
Czech Republic             707.720
Bahrain                    548.400
Saudi Arabia               131.170
Name: Revenue, dtype: float64
```

```
highest_revenue_country = country_revenue.idxmax()
highest_revenue = country_revenue.max()

print("Highest revenue country:", highest_revenue_country)
print("Total revenue:", highest_revenue)
```

```
Highest revenue country: United Kingdom
Total revenue: 8167128.184
```

```
product_revenue = (
    df_clean.groupby("Description")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)

print("Top 10 products by revenue:")
display(product_revenue.head(10))
```

```
Top 10 products by revenue:
```

Description	Revenue
DOTCOM POSTAGE	206245.48
REGENCY CAKESTAND 3 TIER	164459.49
WHITE HANGING HEART T-LIGHT HOLDER	99612.42
PARTY BUNTING	98243.88
JUMBO BAG RED RETROSPOT	92175.79
RABBIT NIGHT LIGHT	66661.63
POSTAGE	66230.64
PAPER CHAIN KIT 50'S CHRISTMAS	63715.24
ASSORTED COLOUR BIRD ORNAMENT	58792.42
CHILLI LIGHTS	53746.66

dtype: float64

```
print("Rows with negative unit prices:")
display(df_clean[df_clean["UnitPrice"] < 0].head())

print("Rows with zero unit prices:")
display(df_clean[df_clean["UnitPrice"] == 0].head())

print("Rows with zero or negative quantities:")
display(df_clean[df_clean["Quantity"] <= 0].head())
```

Rows with negative unit prices:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
299983	A563186	B	Adjust bad debt	1	08/12/2011 14:51:00	-11062.06	NaN	United Kingdom	-11062.06
299984	A563187	B	Adjust bad debt	1	08/12/2011 14:52:00	-11062.06	NaN	United Kingdom	-11062.06

Rows with zero unit prices:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
622	536414	22139	NaN	56	12/01/2010 11:52:00	0.0	NaN	United Kingdom	0.0
1970	536545	21134	NaN	1	12/01/2010 14:32:00	0.0	NaN	United Kingdom	0.0
1971	536546	22145	NaN	1	12/01/2010 14:33:00	0.0	NaN	United Kingdom	0.0
1972	536547	37509	NaN	1	12/01/2010 14:33:00	0.0	NaN	United Kingdom	0.0
1987	536549	85226A	NaN	1	12/01/2010 14:34:00	0.0	NaN	United Kingdom	0.0

Rows with zero or negative quantities:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
141	C536379	D	Discount	-1	12/01/2010 09:41:00	27.50	14527.0	United Kingdom	-27.50
154	C536383	35004C	SET OF 3 COLOURED FLYING DUCKS	-1	12/01/2010 09:49:00	4.65	15311.0	United Kingdom	-4.65
235	C536391	22556	PLASTERS IN TIN CIRCUS PARADE	-12	12/01/2010 10:24:00	1.65	17548.0	United Kingdom	-19.80

```
print("Unique country names before standardization:")
print(df_clean["Country"].unique())
```

```
Unique country names before standardization:
['United Kingdom' 'France' 'Australia' 'Netherlands' 'Germany' 'Norway'
 'EIRE' 'Switzerland' 'Spain' 'Poland' 'Portugal' 'Italy' 'Belgium'
 'Lithuania' 'Japan' 'Iceland' 'Channel Islands' 'Denmark' 'Cyprus'
 'Sweden' 'Austria' 'Israel' 'Finland' 'Bahrain' 'Greece' 'Hong Kong'
 'Singapore' 'Lebanon' 'United Arab Emirates' 'Saudi Arabia'
 'Czech Republic' 'Canada' 'Unspecified' 'Brazil' 'USA'
 'European Community' 'Malta' 'RSA']
```

```
df_clean["Country"] = df_clean["Country"].astype("string").str.strip().str.title()

print("Country names after standardization:")
print(df_clean["Country"].unique())
```

```
Country names after standardization:
<StringArray>
[      'United Kingdom',               'France',            'Australia',
          'Netherlands',              'Germany',               'Norway',
                 'Eire',          'Switzerland',                'Spain',
               'Poland',             'Portugal',                'Italy',
              'Belgium',            'Lithuania',                'Japan',
              'Iceland',      'Channel Islands',              'Denmark',
               'Cyprus',               'Sweden',              'Austria',
               'Israel',              'Finland',              'Bahrain',
               'Greece',            'Hong Kong',            'Singapore',
              'Lebanon', 'United Arab Emirates',         'Saudi Arabia',
       'Czech Republic',               'Canada',          'Unspecified',
               'Brazil',                  'Usa',   'European Community',
                'Malta',                  'Rsa']
Length: 38, dtype: string
```

```
print("Final dataset shape:")
print(df_clean.shape)

print("\nMissing values:")
print(df_clean.isnull().sum())

print("\nDuplicate rows:")
print(df_clean.duplicated().sum())

print("\nFinal dataset preview:")
display(df_clean.head())
```

```
Final dataset shape:
(536641, 9)

Missing values:
InvoiceNo           0
StockCode           0
Description      1454
Quantity            0
InvoiceDate         0
UnitPrice           0
CustomerID     135037
Country             0
Revenue             0
dtype: int64

Duplicate rows:
0

Final dataset preview:
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/01/2010 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	12/01/2010 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/01/2010 08:26:00	2.75	17850.0	United Kingdom	22.00
			KNITTED UNION FLAG HOT		12/01/2010			United	

```
df_clean.to_csv("Online_Retail_Business_Analysis_Cleaned.csv", index=False)

from google.colab import files
files.download("Online_Retail_Business_Analysis_Cleaned.csv")
```